

Episode C2 | *Taxes*

Governments can change your behavior.

How taxes impact markets

This video introduces the intuition for taxes, discusses welfare implications, and uses incidence of tax to motivate elasticity.

Taxes

In the last video we looked at the effect of price controls on welfare.

The second type of government policy we'll look at is taxes.

Governments can tax either sellers or buyers.

If the tax is placed on the seller, the seller is responsible for paying the tax, and this additional cost will be included in the sticker cost of an item.

But when you go to the grocery store, you'll see a price on an item, but a tax gets added on when you check out.

This is a tax on the buyer, since you the buyer are responsible for paying the tax, even though you're paying it through the grocery store.

First, a tax on sellers increases the marginal cost of supplying an item: to get one more unit to the market is more costly, and more costly by exactly the amount of the tax.

So we model this with a higher supply curve to reflect the higher prices.

Then to find prices and quantity, we intersect the taxed supply curve denoted with the dashed line, with the demand curve, to find the quantity.

Then we intersect the quantity with the demand curve to find the price the buyers are paying.

And we intersect the quantity with the untaxed supply curve without to find the seller's price, which is the price the buyers are paying minus the tax.

This works at any reasonable price.

Lets instead place the tax on buyers.

A tax on buyers increases the price faced by buyers.

To buy the same amount, the buyer would need to pay more.

Therefore, for a buyer to be willing to buy the same number of units after the tax, the price without the tax must go down by exactly the amount of the tax.

This demand curve with the tax is denoted with the dashed line.

We find the quantity exchanged by intersecting the demand curve with the tax with the supply curve.

We find the buyers price by intersecting the quantity with the untaxed demand curve.

And we find the sellers price by intersecting the quantity with the supply curve.

It turns out it doesn't matter which side of the market we tax.

The difference between the price paid by the buyer and the price received by the seller is equal to the tax.

So instead of complicating our world with multiple curves with and without taxes, it's simpler to realize that the difference between prices must equal the tax, and that quantity demanded must equal quantity supplied at their respective prices.

And this realization gives us the tax wedge: you can think of sliding the value of the tax in from the left until it gets stuck between the demand and supply curves.

If you want to synthesize your intuition with what we've done in previous videos, you can think of a market without taxes as having the buyer's price and seller's price being equal.

The tax wedge would equal zero and would end up at the equilibrium.

So using our algorithm for finding consumer surplus, we look at the area under the demand curve, above the buyers price, and inside the quantity exchanged.

Producer surplus is the area below the sellers price, above the supply curve, and inside the quantity exchanged.

Government revenue, the revenue the government is receiving from the tax, is just the value of the tax times the number of units taxed.

So this is the area between the buyer's price and the sellers price, inside the quantity exchanged.

And deadweight loss the area between the supply and demand curves above the interval of lost quantity.

Changing taxes changes each of these welfare measures accordingly.

But one important consideration is by how much are prices changing for buyers and sellers?

We call this change in price faced by buyers and sellers, the incidence of tax.

The incidence of tax on buyers, shown in blue, is the difference between the equilibrium price, the grey dashed line, and the buyers price.

And a similar difference for sellers, shown in orange.

And you can see that when we change the relative slopes, the incidence of tax changes as well.

When the demand curve is relatively more elastic, the incidence of tax falls less on buyers.

Buyers are more willing to switch away from the taxed goods, and so face less of the tax.

This result also holds for the supply curve.

When the sellers supply curve is relatively elastic, sellers are more responsive to price, and will bear less of the burden of the tax.

Elasticity

How does the incidence of tax depend on the shapes of supply and demand?

First, way we measure the shapes can't depend on units.

Do the checklist.

Look at how elasticity changes across a range of price quantities: maybe graph these side by side.

Then think about changes in elasticity. Graphically it's going to look like changes in slope. And it's generally true that steeper means less elastic. But remember the reasons we need elasticity instead of slope!